



### Hope

Find out more about the Emirates mission to thoroughly investigate the atmosphere of the Red Planet. <https://digit.in/jul20-77>



### Mars 2020

More info on NASA's Mars 2020 mission, including objectives, instruments on board Perseverance. <https://digit.in/jul20-78>

# The 2020 launch window

**We will see a number of probes heading out to Mars next month**

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bout every two years or so a launch window to Mars opens up, which is when a probe can

be sent to the Red Planet from Earth using the least amount of fuel. The window remains open for a few weeks and since 2011, humans have been using every possible opportunity to send a mission to Mars. The Fobos-Grunt mission was sent in 2011, then the MAVEN and Mars Orbiter Mission (Mangalyaan) in 2013, followed by the ExoMars mission along with the Schiaparelli lander in 2016, and the InSight mission in 2018. The next opportunity for launch opens up on July 14 and will last till around August 11. There are no less than three missions planned to make the most out of the 2020 launch window.

### HOPE

Hope is an Emirati orbiter that is scheduled for launch on a Japanese Mitsubishi HIIA rocket whenever a suitable opportunity opens up after mid July. This is the first mission to the Red Planet by the United Arab Emirates. The spacecraft will spend 200 days cruising, and will reach its target orbit around Mars in May 2021. The primary goal of the mission is to characterize the atmosphere of Mars, by monitoring

it continuously for two years, with a possible planned extension of another two years.

The orbiter will study the dynamics of the upper atmosphere of Mars, something that the MoM mission did as well. In particular it will look at the behavior of oxygen and hydrogen in the atmosphere. The constant action of the Solar wind is stripping away the already thin atmosphere on Mars, and the Hope mission is expected to find out more about the complex interactions between the Martian atmosphere, the Solar wind and outer space. Hope will also investigate any connections between the current weather on Mars, and the possible climate conditions in the past. There is expected to be some kind of complex interaction between the lower and upper atmosphere of Mars, and Hope will investigate this as well. When it comes to Methane, there is a mysterious cycle where Methane levels seem to spike in daily as well as yearly intervals. Some of these spikes have been measured by other orbiters, and even by rovers on the ground. The studies by Hope will allow scientists to better understand the complex atmosphere on Mars. Hope will study how the topography of the planet affects local weather conditions, monitor global changes in temperature, and observe weather events such as dust storms. Hope will be able to provide a planet wide record of how the Martian atmosphere changes across a single day, and over the course of a year as well.

The payloads on board to support these observations are The Emirates eXploration Imager (EXI) which is a camera that can capture images in both the visible spectrum and UV light, the Emirates Mars Ultraviolet Spectrometer (EMUS), a spectrograph in the far UV range and the Emirates Mars InfraRed Spectrometer (EMIRS). The spacecraft can beam back data at a rate of 1.6 Mbps.

### MARS 2020

The Mars 2020 mission is an ambitious one by NASA that hopes to explore the conditions of Mars really long ago to check if they were suitable for life, and to directly identify any signs of life. The main component of the mission is the Perseverance rover, which is scheduled for launch on a suitable opportunity after July 14. The launch is slated to take place on the brand new rocket by the United Launch Alliance, the Atlas 541. It will reach Mars in February 2021, the chosen landing site being an ancient river delta known as the Jezero Crater. The landing on the surface is similar to the curiosity rover, and will use a parachute in tandem with a sky crane, which is a frame with four reverse thrusters that will gently lower perseverance on the surface. This one has a microphone on board which will provide the scientists with additional information during the descent, and allow NASA to identify any problems during the regular operations of the rover. It is a staggered sample collection